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ORIGINAL PAPER

Application of a hybrid Taguchi grey approach for determining the optimal parameters on laser beam welding of dissimilar metals

P. Thejasree[1] · Manikandan Natarajan[1] · Muhammed Anaz Khan[2] · Sai Vempati[3] · Balram Yelamasetti[2] · Abhishek Dasore[4]

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Abstract
There has been a consistent increase in the utilization of laser sources, resulting in a decrease in expenses and an increase in efficiency. Several factors have contributed to this phenomenon, including advancements in procedures and machinery such as laser beam welding systems. Inconel 718 is a highly utilized material in the manufacturing of chemical, nuclear, and marine equipment. Conversely, SS304 is commonly used in the food processing, automotive, and aerospace industries. It is crucial to conduct a thorough investigation of the welding techniques employed for dissimilar metals, with particular emphasis on SS304 and Inconel 718. The current investigation employed the Taguchi design methodology. The amalgamations of LP at 2.7 kW, WS at 2.35 mm/min and PD at 6.5 ms accomplishes minimalized top and bottom width. Likewise, an amalgamation with 3.1 kW LP, 2.05 mm/min WS and 7.5 ms PD accomplishes maximized feasible penetration which consequences in quality joints. The research has revealed that the laser power is the primary process variable that significantly impacts the width of both the upper and lower sections. The velocity at which welding is performed is a crucial factor that influences the rate of penetration. The attainment of optimal penetration can be facilitated by the combination of elevated levels of peak power and pulse duration with reduced levels of weld speed.

Keywords Laser welding · Dissimilar welding · Inconel 718 · SS304 · Taguchi approach · Grey approach · Contour analysis · Interaction analysis

1 Introduction

Metal welding is a process that can be used to efficiently utilize different materials. It allows a product to be adaptable and profitable by taking advantage of the various properties of each material. It is commonly utilized in various applications such as oil processing plants, power plants, and chemical and petrochemical facilities [1–3]. Due to the checking related point of view and leading unreachable working conditions, subsea applications necessitate highly reliable equipment. The dependability of the approach and materials adopted to manufacture components is critical for ensuring the equipment's long-term integrity [4]. Due to the increasing number of applications of stainless steel in various industries, countries are looking for corrosion-resistant materials. Among the various types of stainless steel that are generally adopted in the making of marine equipment is duplex stainless steel (DSS). Duplex stainless-steel grades are normally engaged in the construction of various industrial facilities such as pipelines, and concrete constructions for oil and gas pipelines. Due to the contribution of nitrogen in the chemical constituents of material, the alloying methods for contemporary DSS became widely used in 1980 [5–7]. Amidst of available DSS types, super DSS 2507 gaining popularity and its contribution for numerous engineering uses has grown over the last few decades. The necessary mechanical characteristics of super DSS are better than remaining grades and provides economic benefits without convincing

✉ Balram Yelamasetti
Balram3072@gmail.com

1 Department of Mechanical Engineering, School of Engineering and Technology, Mohan Babu University, Tirupati, Andhra Pradesh, India

2 Department of Mechanical Engineering, MLR Institute of Technology, Hyderabad 500043, India

3 University of A Coruña, CIF, CITENI, A Coruña, Spain

4 Department of Mechanical Engineering, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Andhra Pradesh 522302, India

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the properties [8–10]. Dual-phase super DSS 2507 has ferritic and austenitic stainless-steel properties. Dual-phase super DSS 2507 outperforms other DSS 2205 grades in corrosion resistance [11]. Because super DSS 2507 has excellent mechanical and corrosive behaviour, it is widely adopted in especially in corrosive environments [12]. Nitronic steel possess better weldability with better mechanical behaviour and corrosive behaviour than commercially available SS grades and it is commonly employed in marine uses and other components [13–16].

Inconel 625, a Nickel alloy, is the material with significant uses in the aviation, aeronautics, and other corrosion environments. It possesses expected mechanical characteristics such as strength and corrosive resistance in severe environments. 'Ni' alloys are commonly used in numerous engineering uses and also in the academic community [17–19]. Dissimilar welding of SS and 'Ni' alloys have been adopted especially in corrosive environments. Despite the various advancements that have occurred in the field of nickel and stainless over the past couple of decades, most of the reviews that have been written about these materials have not covered the joint of DSS 2507 with austenitic superalloys and nickel. This review concentrates on the joint of super DSS 2507, which is composed of austenitic austenitic steel, Inconel 625, and superalloy. It analyzed the various mechanical and structural properties of the joint using different GTAW processes. The study also focused on the impact of filler compositions, residual stress deformation, and heat input on the joint's performance. Due to the stable austenitic condition of the Ni–Cr matrix, Inconel 625 alloy is commonly used in high and low temperature environments. Its chemical and physical properties make it an ideal choice for welding applications [20, 21]. The mechanical strength of certain types of materials, such as those used in marine, aeronautical, and petrochemical industries, is usually improved by the use of Ni-base alloy. These include components for power plants, boiler systems, and turbine blades for aircraft [22–24]. In order to create dendritic and defect-free weldments, the researchers utilized laser beam welding. They noted that the various interfaces of the joints were not contaminated with harmful intermetallics. This technique is a promising method for producing high-strength joints. Despite its growing popularity in the industrial sector, there is a lack of research regarding its various uses [25–28]. The goal of the project is to use beam-based weld approach for joining various superalloy materials, such as those utilized in the production of fuel injection parts. This method is commonly used in the automotive industry to produce joint structures between two sheets superalloy. In addition to being faster than traditional methods, the beam welding process also utilizes a deep penetration and small HAZ to produce high-quality joints [29–32]. The laser beam welding process is ideal for creating efficient and effective weldments between various metals. The researchers intend to use this technique to join superalloy materials, such as SS304 and Inconel 718, which are based on Ni.

From the available literature, there is a need of performing investigations on laser beam welded dissimilar joints which is a defect-free joint of 1 mm thick with better penetration. The optimization of LBW dissimilar joints made of Inconel 718 and SS 304 was performed through interaction and contour plots.

2 Materials and methods

The purpose of this research is to look into the efficacy of a laser welding method called as LBW in making of dissimilar weldments. The work specimen adopted in the analysis are 1 mm thick Inconel 718 and SS304 sheets as represented in Fig. 1.

For preparation of weldments, a 4-kW pulsed Nd-YAG laser setup, has been adopted. With help of proper arrangements, the work specimen have been held in the setup. The continuous pulse of the machine confirms that the components are allied flawlessly. The process parameters considered in this analysis are represented in Table 1. The schematic of weld geometry measurement has been represented in Fig. 2.

3 Result and discussion

The experimentations were performed using the L27 orthogonal array. The aspiration of the exploration is to enhance the welding performance of two dissimilar materials. The lower the width of the top or bottom portion, the better the criterion's effectiveness.

3.1 Ascendency of factors on top width

Figure 3 portrays the response plot for top side width achieved for LBW of dissimilar metals. The top width during LBW shows an increase in its width as the pulse duration and laser power get longer. This is because the higher the power, the more amount of heat is exposed to the metal. As the duration of the pulse gets longer, the number of cycles that cause heat to the metal also increases. The speed at which the weld is conducted also affects the duration of the dwell period of the supplied heat to the metal. As a result, the beads in the weld are narrower at optimum speeds [17, 27].

The response analysis has been represented in Table 2 and it is noticed from the analysis that the amalgamation of process factors (A1B3C1) produces minimum top width which means LP at 2.7Kw, WS at 2.35 mm/min and PD at 6.5 ms. According to the study, laser power is the most influential variable during LBW of dissimilar welding.

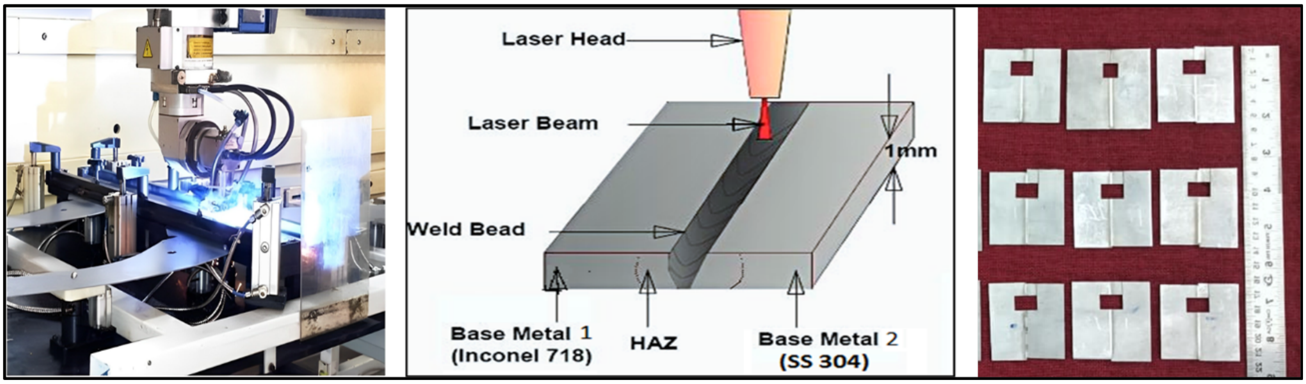

Fig. 1 Schematic of laser welding (Inconel 718 and SS304)

Table 1 Process parameters during laser welding and levels

Symbols	Weld Parameters	Levels		
		1	2	3
A	Laser Power (kW)	2.7	2.9	3.1
B	Weld Speed (mm/min)	2.05	2.2	2.35
C	Pulse Duration (ms)	6.5	7.0	7.5

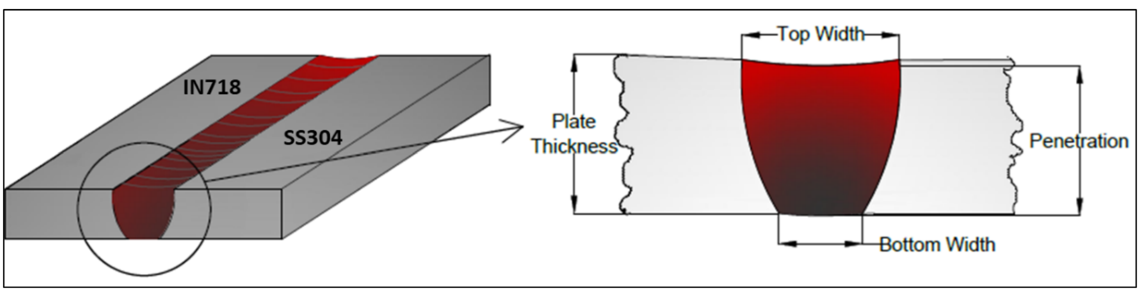

Fig. 2 Schematic of weld geometry measurement

3.2 Ascendency of factors on bottom width

The graph showing the response of the bottom width of the welded joint using LBW is presented in Fig. 4.

The bottom width of the joint is becoming more prominent with the increasing pulse duration and laser power. On the other hand, its decreasing trend is related to the increasing weld speed. Because of lower heat input from the top side to the bottom side, the width of the joint that's measured is smaller than that of the top [17, 27].

The response analysis has been shown in Table 3 and it is noticed from the analysis that the amalgamation of process factors (A1B3C1) which means which means LP at 2.7Kw, WS at 2.35 mm/min and PD at 6.5 ms produces minimum

Table 2 Response table for top width

Levels	Means of responses			S/N ratio		
	A	B	C	A	B	C
1	**1.599**	1.806	**1.698**	– 4.117	– 5.279	– 4.684
2	1.751	1.777	1.752	– 4.943	– 4.996	– 4.948
3	2.022	**1.758**	1.901	– 6.083	– 4.868	– 5.510
Delta	0.423	0.048	0.203	1.966	0.411	0.826
Rank	1	3	2	1	3	2

Bold values are more significant at particular level

Table 3 Response table for bottom Width

Levels	Means of responses			S/N ratio		
	A	B	C	A	B	C
1	**0.1502**	0.4019	**0.2487**	17.498	10.821	12.936
2	0.3689	0.3699	0.3715	9.081	10.347	10.027
3	0.5887	**0.3368**	0.4882	5.420	10.830	9.036
Delta	0.4385	0.0651	0.2395	12.078	0.482	3.900
Rank	1	3	2	1	3	2

Bold values are more significant at particular level

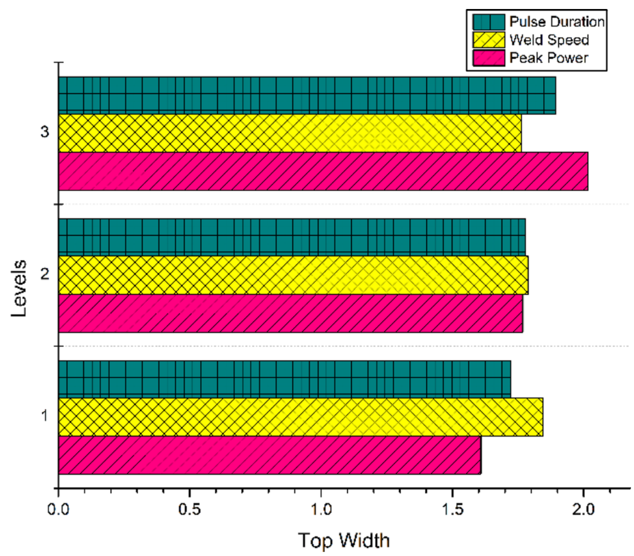

Fig. 3 Response plot for top width

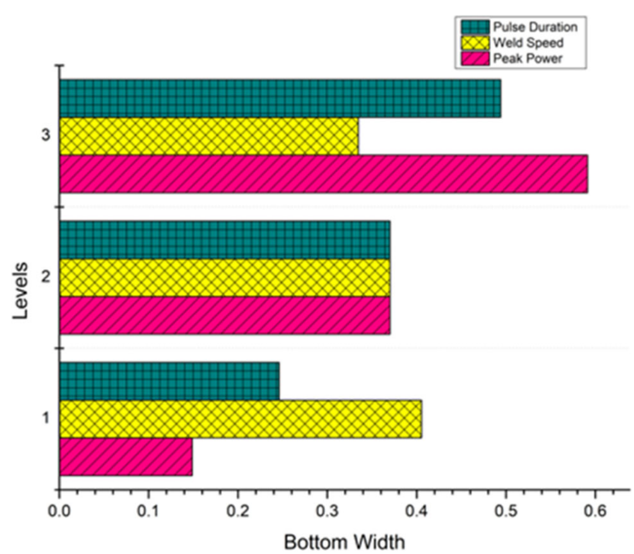

Fig. 4 Response plot for bottom width

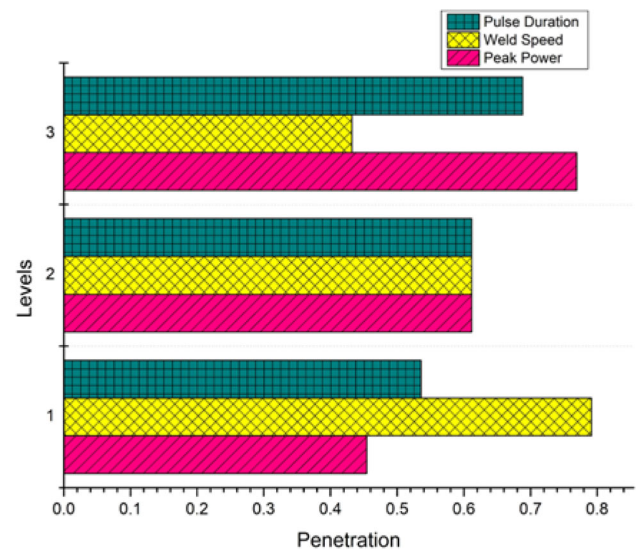

Fig. 5 Response plot for penetration

bottom width. According to the study, laser power is the most influential variable during LBW of dissimilar welding.

3.3 Ascendency of factors on penetration

The graph showing the penetration rate achieved by the LBW process dissimilar weldments is presented in Fig. 5.

The graph shows the increasing penetration rate achieved by laser beam welding due to the increasing pulse duration and power. It's also related to the decreasing penetration rate as the welding speed increases. This occurs because the faster the weld speed, the less duration it takes for the heat to reach the part that's being fused.

The Table 4 containing the response data for the LBW process on dissimilar metal joints has been presented. It shows that the amalgamation of process variables A3B1C3 which means the combinations of 3.1 kW, WS2.05 mm/min, PD 7.5 ms improved its penetration rate. The investigators noted that the speed at which the weld is conducted is the most influential factor in defining the joint's penetration rate.

Table 4 Response table for penetration

Levels	Means of responses			S/N ratio		
	A	B	C	A	B	C
1	0.4603	**0.8009**	0.5411	− 10.173	− 2.115	− 8.333
2	0.6132	0.6155	0.6224	− 4.573	− 4.618	− 5.067
3	**0.7700**	0.4352	**0.6798**	− 2.303	− 10.317	− 3.649
Delta	0.3097	0.3657	0.1387	7.871	8.202	4.684
Rank	2	1	3	2	1	3

Bold values are more significant at particular level

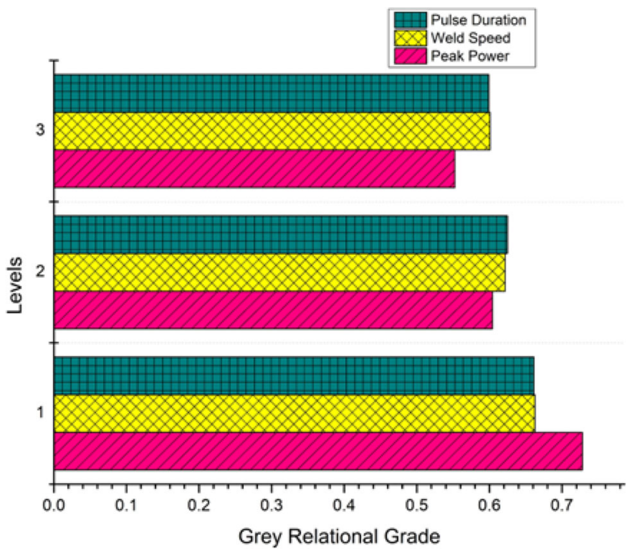

Fig. 6 Response plot for GRG

3.4 Ascendency of factors on GRG

The response graph for GRG is represented in Fig. 6. The multi performance welding has been amplified with an increment of pulse duration, peak power and weld speed. It is noted that peak power is the most influencing process variable that influences overall welding performance.

4 Contour plot analysis

Contour plot analysis is the 2D depiction of three different factors. Figure 7a–c epitomizes the contour analysis for top width and it is perceived that the combinations of lesser pulse duration and peak power offers minimum top width. Similarly, the combinations of higher weld speed and lower peak power, lower pulse duration and higher weld speed offers minimum top width. Figure 7d–f illustrates the contour analysis for bottom width and it has the similar trend like top width.

Figure 8a–c represents the contour analysis for penetration during LBW of dissimilar metals. It is perceived from the depiction that the combinations of higher pulse duration and peak power offers improved penetration. Similarly, the combinations of lower weld speed and higher peak power, higher pulse duration and lower weld speed offers better penetration.

Figure 8d-f illustrates the contour analysis for GRG which is expected to be maximum the better. It is noticed from the analysis that the combinations of lower pulse duration and peak power offers better and improved multi aspects welding performance. Similarly, the combinations of lower weld speed and peak power, lower pulse duration and weld speed produces the better multi performance welding

5 Interaction plot analysis

Interaction analysis is adopted to reveal the interaction effect of input variables on dependent variables. Figure 9 shows the interaction plot for top width. It is noticed from the analysis the weld speed and peak power; weld speed and pulse duration have the considerable amount interaction on top width. Figure 10 depicts the interaction plot for bottom width. It is perceived that the selected independent variables at all levels have predominant interaction bottom width. The interaction analysis for penetration has been done and plotted in Fig. 11. It is obvious that the factors at different levels have preeminent interaction on penetration. The interaction analysis for GRG is represented in Fig. 12. All the independent factors with all the levels have significant interaction effect on multi performance index.

6 Conclusions

The study of laser welding processes and the determination of parameters for achieving better joint performance in LBW of dissimilar metal are crucial for various manufacturing operations. This investigation was carried out using Taguchi-Grey

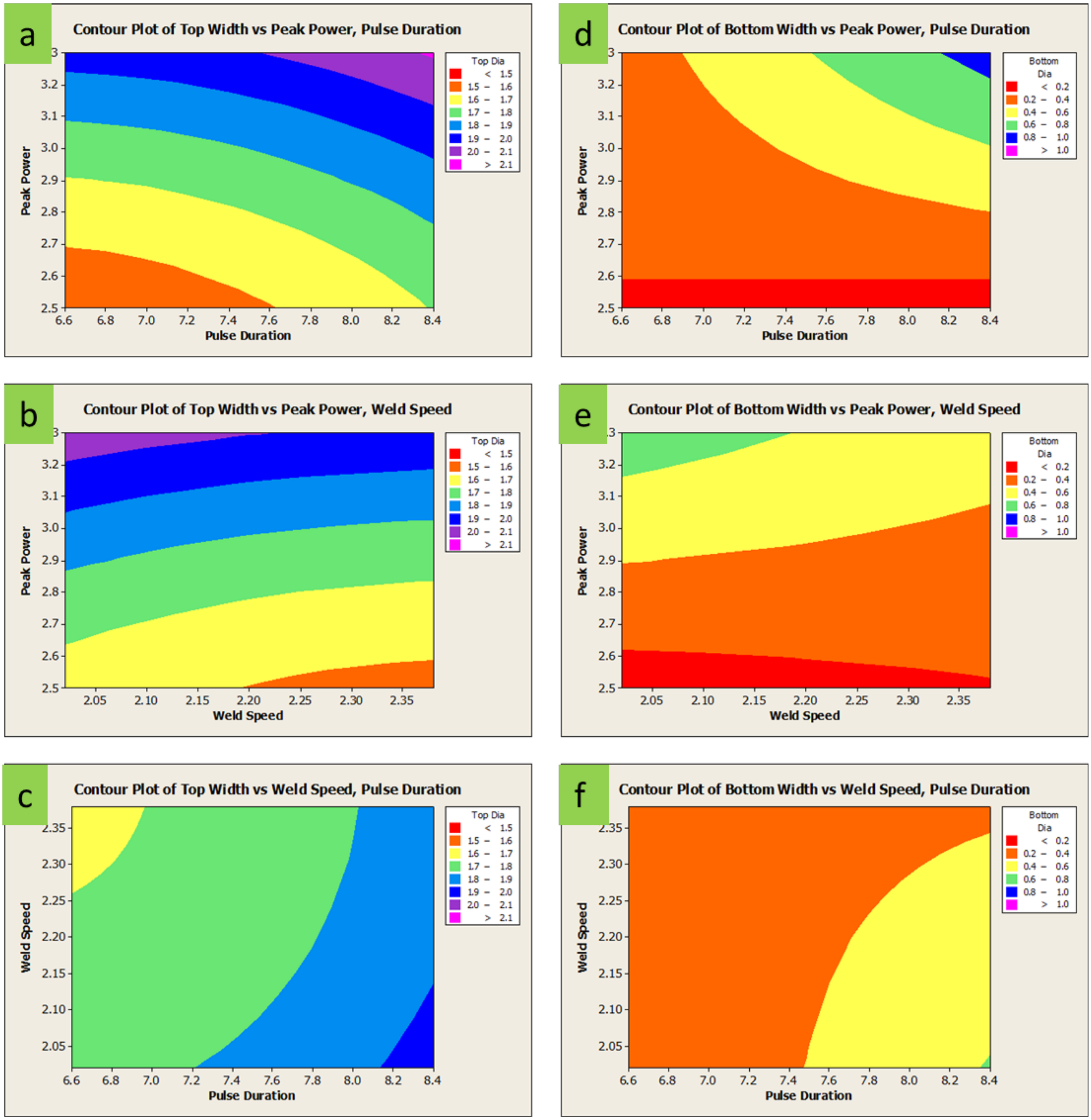

Fig. 7 Contour analysis for top and bottom width

approach. The outcomes of the investigational analysis have been presented as follow:

- The experimentation have been done as per the design approach of Taguchi, which involves taking into account the input factors for the LBW process of Inconel 718 and SS304 sheets. The response analysis determined the optimal parameters for the improvement of the welding performance.
- It has been concluded that the power of the laser is the most important process variable that influences the width of the top and bottom sections. The weld speed also has a significant effect on the penetration rate.
- The outcomes of the exploration divulged that the combination of the pulse duration, weld speed, and laser power at different levels can achieve the minimum top and bottom width.

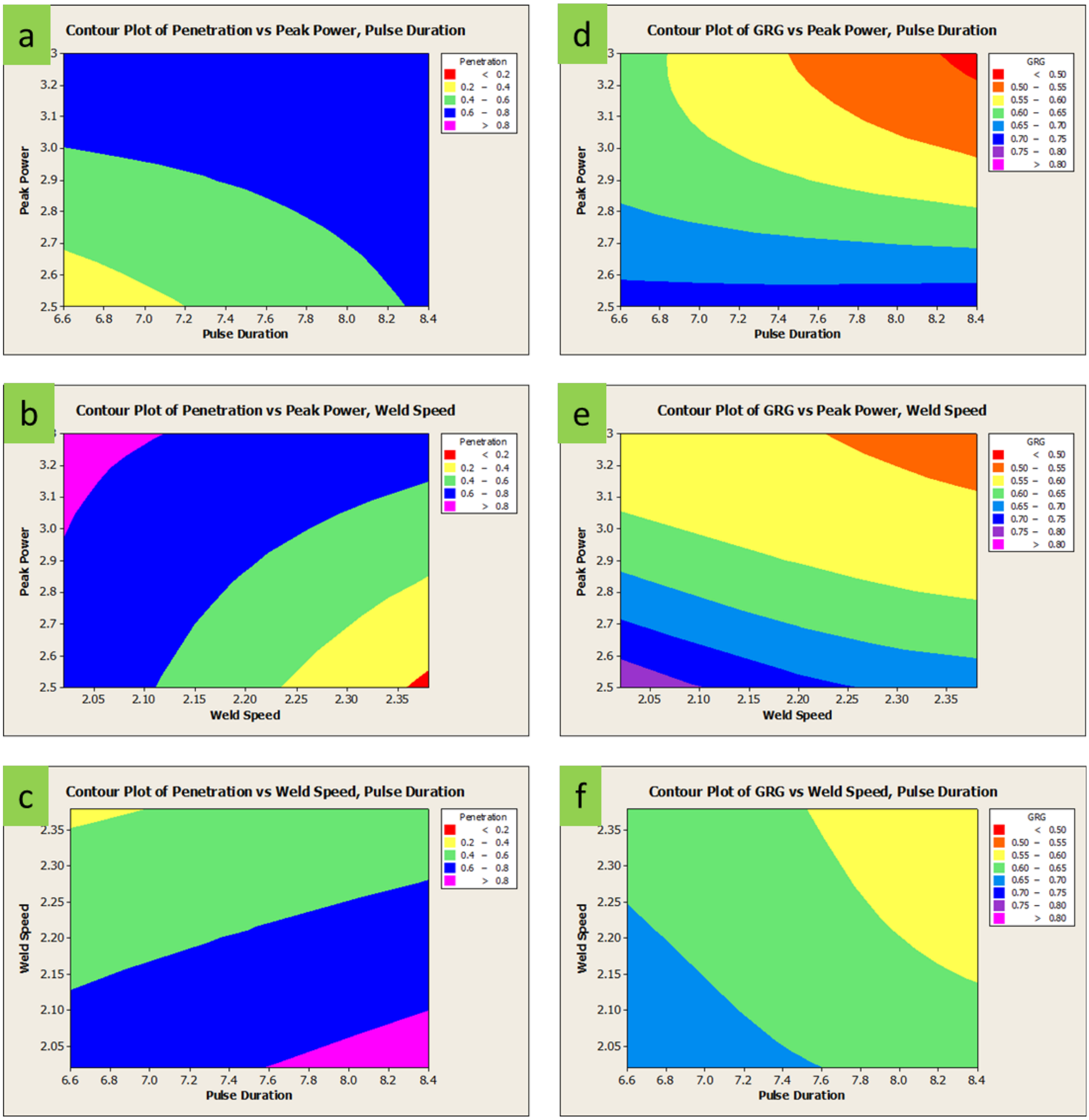

Fig. 8 Contour analysis for penetration and GRG

- The amalgamations of LP at 2.7 kW, WS at 2.35 mm/min and PD at 6.5 ms accomplishes minimalized top and bottom width. Likewise, an amalgamation with 3.1 kW LP, 2.05 mm/min WS and 7.5 ms PD accomplishes maximized feasible penetration which consequences in quality joints.
- Grey approach has been espoused to establish the process parameter combination for achieving the multi performance welding.
- Contour and interaction analysis also executed to divulge the combinatory and interaction impact of various parameters on chosen output characteristics.
- The suggested model may be utilized for different joining applications. It can be particularly beneficial when carrying out advanced joint fabrications.

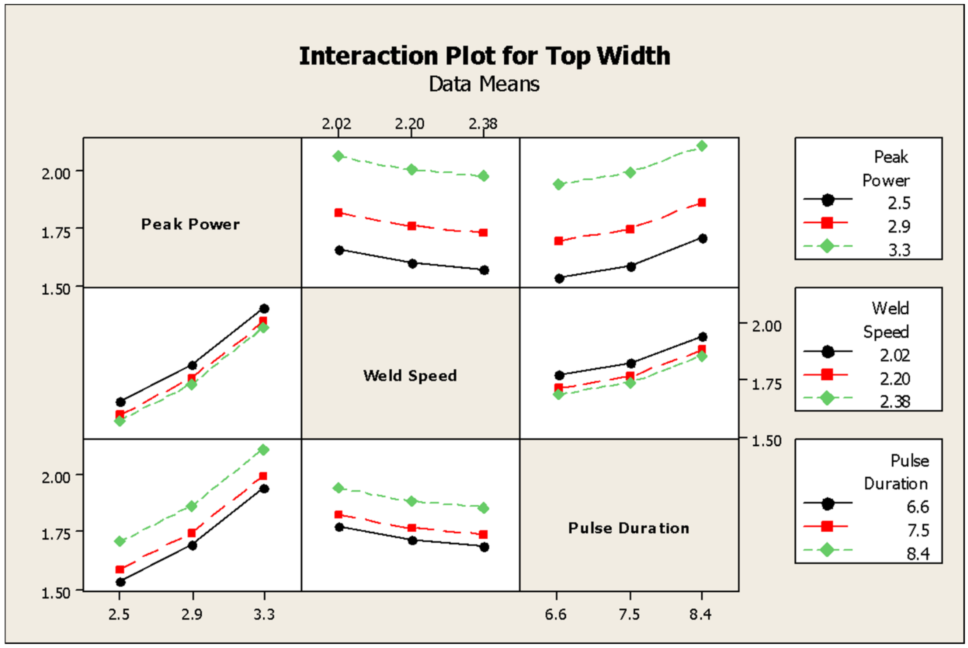

Fig. 9 Interaction analysis for top width

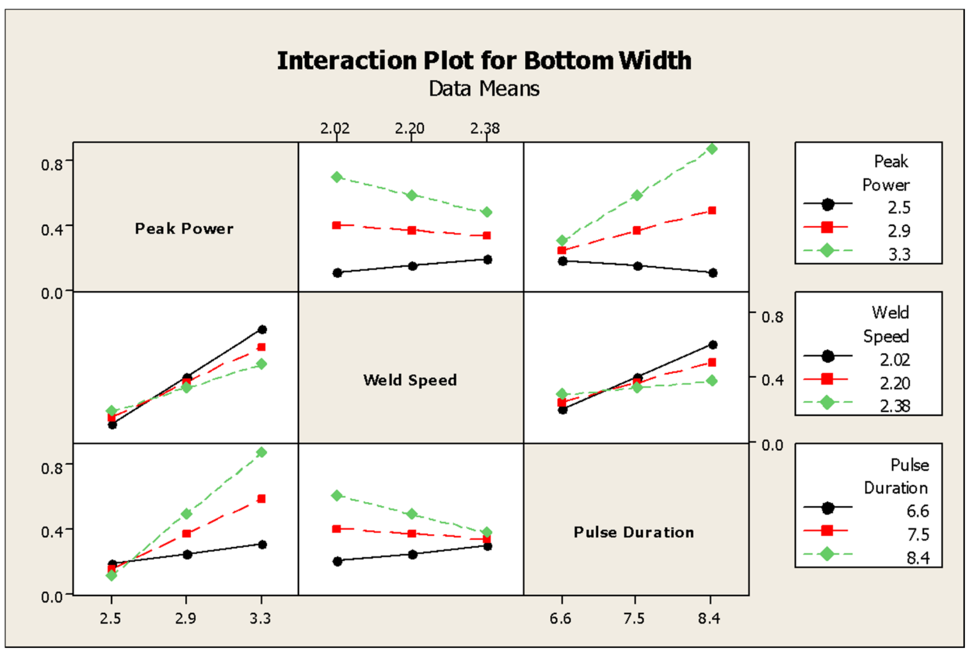

Fig. 10 Interaction analysis for bottom width

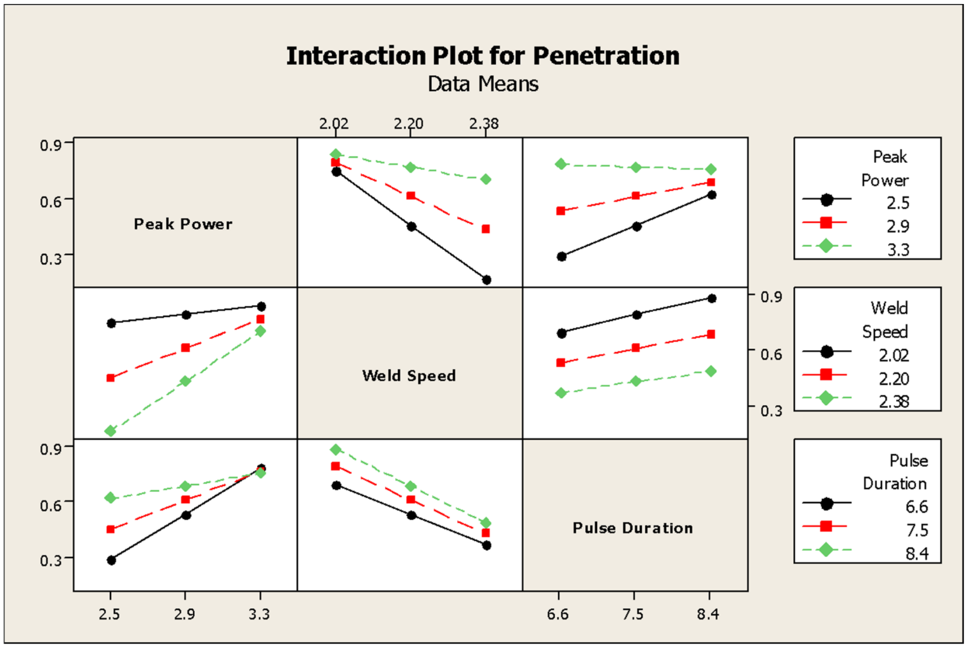

Fig. 11 Interaction analysis for penetration

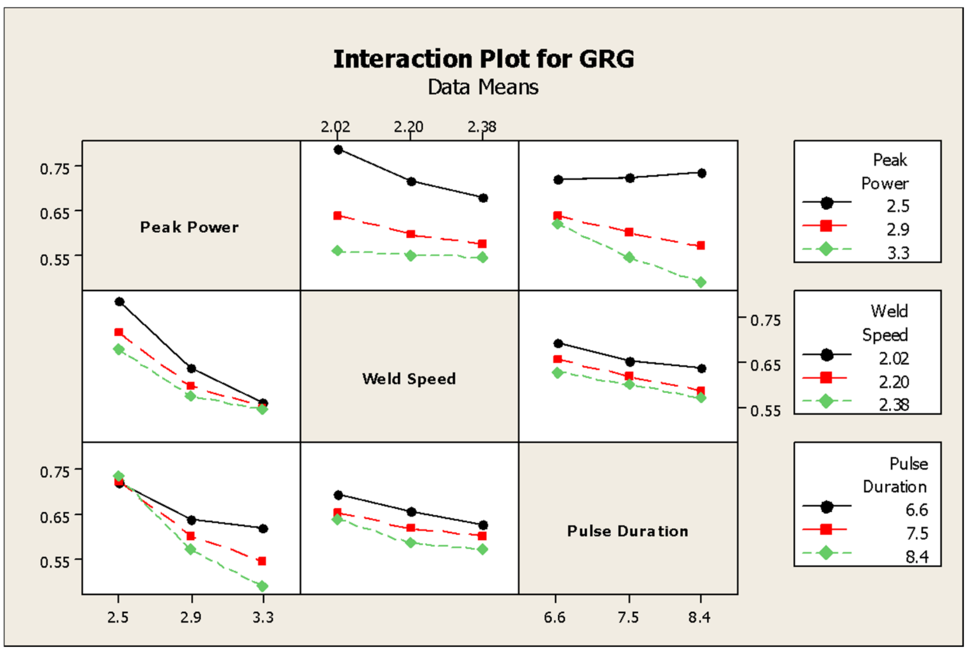

Fig. 12 Interaction analysis for GRG

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